

**CS-413 Software Engineering
(Mid Semester Test I)**

Date: October 3, 2024

Time: 1 hr 15 Min.

Marks: 20

Answer all the questions. Each question carries 5 marks.

1. Why does an iterative process make it easier to manage change? Is it possible to complete a project in just one iteration and still be agile? Explain your answer.
2. Develop a Use Case diagram for Hotel Management System and write Use Scenario for room booking in a hotel.
3. Develop a set of CRC (Class Responsibility Collaborator) cards for Library Management System.
4. Discuss the relationship between the concept of information hiding as an attribute of effective modularity and the concept of module independence.

**School of Computer and Systems Sciences
Jawaharlal Nehru University, New Delhi**

**End Semester Examination,
Monsoon Semester 2024
CS 413—Software Engineering**

Time: 3 Hrs.

Date: December 17, 2024

Marks: 60

Note: Answer all the questions. Each question carries (6+6=12) marks.

1. (a) What do you understand by the Spiral style of software development? Graphically depict the activities that a programmer typically carries out while developing a programming solution using the Spiral style. In your diagram also show the order in which the activities are carried out.

(b) What is the basic difference between a control flow-oriented and a data flow-oriented design technique? Can you think of any reason as to why a data flow-oriented design technique is likely to produce better designs than a control flow-oriented design technique?
2. (a) Explain the Model-View architecture style using Smart Home as example.

(b) Describe (I) the worst interface that you have ever worked with and (II) the best interface that you have ever worked with and critique both of them with standard interface design concepts.
3. (a) What is Software refactoring? Consider a case of basic School Management System and explain the refactoring concepts through class diagram (I) before refactoring (II) after refactoring is done.

(b) What is the objective of software maintenance? Explain the basic difference between preventive, corrective, adaptive and perfective software maintenance.
4. (a) What is the Black box testing? Discuss any black box technique with an example.

(b) How a control flow diagram can help in White box testing.
5. (a) What are the different project parameters that determine the cost of a project? What are the important factors which make it hard to accurately estimate the cost of software projects? If you are a project manager bidding for a product development to a customer, would you quote the cost estimated using COCOMO as the price in your bid? Explain your answer.

(b) Discuss the empirical, heuristic and analytical estimation techniques in Software engineering.

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4. (a) What is the Cause-effect Graph based Black box testing? Draw the Cause-Effect graph for a given scenario and discuss the possible test cases.

The "Print Message" is software that reads two characters and depending on their values, messages are printed.

- *The first character must be an "A" or "B".*
- *The second character must be a digit.*
- *If the first character is an "A" or "B" and the second character is a digit, then the file must be updated.*
- *If the first character is incorrect (not an "A" or "B"), the message X must be printed.*
- *If the second character is incorrect (not a digit), the message Y must be printed.*

(b) Draw the control flow diagram and possible test paths for the following piece of code.

```
int binsearch(int x, int v[], int n)
{
    int low, high, mid;
    low=0;
    high= n-1;
    while(low<= high)
    {
        mid= (low + high)/2;
        if (x < v[mid])
            high= mid - 1;
        else if (x > v[mid])
            low = mid + 1;
        else return mid;
    }
    return -1;
}
```

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- (b) Discuss the empirical, heuristic and analytical estimation techniques in Software engineering.